Project Plan

Customer Support RAG Chatbot

Project Type: Depi Graduation Project

Domain: AI-Powered E-Commerce Customer Support

Duration: 11 Weeks

Team Size: 5 Members

1 Timeline

The project follows an 11-week development cycle divided into 9 overlapping phases. The Gantt chart below illustrates the schedule and task dependencies.

PHASE	START	END	DURATION	MILESTONE
Data Collection & Exploration	Week 1	Week 2	2 weeks	Data Collection & Cleaning
Data Cleaning & Preprocessing	Week 2	Week 3	1.5 weeks	Data Collection & Cleaning
NLP & Embedding Pipeline	Week 3	Week 5	2 weeks	Retrieval
Backend API Development	Week 4	Week 6	3 weeks	Backend
RAG Pipeline Integration	Week 5	Week 7	2 weeks	Generation
Authentication & Security	Week 7	Week 8	1.5 weeks	Backend
MLOps & Monitoring	Week 8	Week 9	1.5 weeks	MLOps
Adaptive RAG & Testing	Week 9	Week 10	1.5 weeks	Generation
Documentation & Deployment	Week 10	Week 11	1 week	Frontend

2 Milestones & Deliverables

The project is organized around 6 core milestones. Each milestone produces specific deliverables that feed into subsequent phases.

#	MILESTONE	TIMELINE	DELIVERABLES
M1	**Data Collection & Cleaning**	Week 1 – Week 3	• Raw dataset (BrownBox e-commerce Q&A — 46,924 records) • Cleaned dataset (`cleaned_data.csv`) • QA chunks dataset (`qa_chunks.csv`) • Exploratory Data Analysis (EDA) notebooks • Data cleaning & preprocessing notebooks
M2	**Retrieval**	Week 3 – Week 5	• Sentence-Transformers embedding pipeline (`all-MiniLM-L6-v2`) • ChromaDB vector database with ingested FAQ data • Retriever module with confidence scoring (`1 - distance`) • Data ingestion script (`ingest_data.py`) • Embedding experiments notebook
M3	**Generation**	Week 5 – Week 10	• LLM Generator with 4 methods (generate, rewrite, verify, direct answer) • Adaptive RAG pipeline with 5 confidence paths • LLM Intent Router (CASUAL vs SUPPORT classification) • Query rewriting for vague follow-up questions • Answer verification layer for low-confidence responses • Pipeline benchmark test suite with MLflow logging
M4	**Backend**	Week 4 – Week 8	• FastAPI REST API with structured endpoints • PostgreSQL database with 5 ORM models (User, Conversation, ChatMessage, Feedback, OTPCode) • Google OAuth 2.0 + JWT authentication + OTP password reset • Rate limiting (SlowAPI) for critical endpoints • Conversation history & memory management • Voice input endpoint with Whisper speech-to-text • Feedback system (thumbs up/down with comments) • Structured logging middleware with unique request IDs
M5	**Frontend**	Week 10 – Week 11	• Next.js web application • Chat interface with conversation management • User authentication UI (login, register, OAuth) • Responsive design and deployment
M6	**MLOps**	Week 8 – Week 9	• MLflow experiment tracking (logs every chat query) • Prometheus metrics (counters, histograms, gauges) • Streamlit monitoring dashboard with 5 KPIs and 4 charts • Configuration comparison experiments (`mlflow_experiment.py`) • `/metrics` endpoint for Prometheus scraping

3 Resource Allocation

The following resources were allocated to support the development, deployment, and monitoring of the project.

TEAM SIZE

5

Team Members

CLOUD / APIS

- Groq API (LLM Inference)
- Google OAuth 2.0 (Authentication)
- Gmail SMTP (OTP Email Delivery)

SOFTWARE STACK

- Python (Core Language)
- FastAPI (Backend Framework)
- PostgreSQL (Relational Database)
- ChromaDB (Vector Database)
- MLflow (Experiment Tracking)
- Next.js (Frontend Framework)

AI / ML MODELS

- Llama 3.3 70B Versatile — via Groq (LLM)
- all-MiniLM-L6-v2 — Sentence-Transformers (Embeddings)
- Whisper Small — OpenAI (Speech-to-Text)